Lab 1 – Requirements Table

Problem Statement #28 – Inter-City Freight Load Matching Marketplace

Domain: Smart Cities, Transport & Logistics | Actors: Shipper, Freight Carrier, Carrier Verification Service, Escrow Payment Gateway

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale
FR-001	Functional	The system shall allow a verified trucking carrier to submit a competitive bid (rate, equipment type and pickup ETA) on a posted freight load, only while the shipper-defined auction window is open.	High	Pass: A carrier with valid permits, insurance and KYC status submits a bid before the auction close time; the bid is stored with a server timestamp and the shipper is notified within 5 seconds. Fail: A bid is accepted from an unverified carrier, or after the auction close time has passed.	Competitive bidding is the core price-discovery mechanism of the marketplace; permit checks keep illegal carriers off the exchange.
FR-002	Functional	The system shall allow a shipper to post a freight load specifying pickup and drop-off locations, pickup/delivery windows, cargo type, weight/volume, required vehicle type, reserve price and auction close time.	High	Pass: A load with all mandatory fields completed is published with a unique Load ID and becomes visible to matching carriers within 10 seconds. Fail: The load is saved with any mandatory field blank, or with a pickup date earlier than the current date.	Loads are the primary supply-side input; complete, structured data is required for carrier matching and for a valid freight contract.
FR-003	Functional	The system shall allow a shipper to review ranked bids and award the load to one carrier, upon which the platform shall lock the winning bid, generate a digital freight contract and place the agreed amount in escrow.	High	Pass: On award, load status changes to 'Awarded', a contract document is generated, an escrow hold equal to the bid value is created, and both parties are notified. Fail: A load is awarded to a carrier whose verification has expired, or the same load is awarded to two carriers.	Converts a winning bid into a binding, funded commitment and prevents double-allocation of the same shipment.
FR-004	Functional	The system shall allow the assigned carrier to submit electronic proof of delivery (consignee e-signature, geo-tagged timestamp and scanned delivery note) and shall validate it before marking the delivery milestone complete.	High	Pass: A POD containing a consignee signature and GPS coordinates within 500 m of the registered drop-off point is accepted and the milestone is set to 'Delivered'. Fail: A POD is accepted without a signature, or with coordinates outside the delivery geofence.	Verified POD is the trigger for automatic escrow release, so its authenticity directly protects both shipper funds and carrier payment.

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale
FR-005	Functional	The system shall record shipment milestone updates (Assigned, In-Transit, At-Destination, Delivered) and display the current status to the shipper and the carrier, notifying both parties on every status change.	Medium	Pass: A milestone update is reflected on both dashboards and a push/e-mail notification is delivered within 30 seconds of the update. Fail: An update takes longer than 30 seconds, or shipment status is visible to a user not party to that contract.	Visibility reduces check-call overhead for inter-city freight and provides the audit trail used to settle delivery disputes.
NFR-001	Non-functional (Performance & Security)	Milestone payments held in escrow shall be disbursed automatically to the carrier within 60 seconds of a verified electronic proof-of-delivery submission, over an encrypted and fully audit-logged transaction channel.	High	Pass: Benchmarking with 1,000 concurrent POD submissions shows 95th-percentile disbursement time under 60 seconds, all transactions carried over TLS 1.3 and every disbursement written to an immutable audit log. Fail: p95 exceeds 60 seconds, or any disbursement is missing from the audit log.	Fast, tamper-evident settlement is the platform's key differentiator and directly affects carrier cash flow and trust.
NFR-002	Non-functional (Reliability & Scalability)	The platform shall maintain at least 99.9% monthly availability and shall support 5,000 concurrently open auctions, with 95th-percentile API response time not exceeding 2 seconds.	High	Pass: A load test of 5,000 concurrent auctions and 10,000 bids per hour records p95 response time of 2 seconds or less, and the monthly uptime report shows availability of 99.9% or higher. Fail: p95 exceeds 2 seconds under the load test, or monthly downtime exceeds 43 minutes.	Auctions are time-boxed, so an outage or slow response during the closing minutes of bidding causes direct revenue loss for both sides.

Traceability to Use Cases

FR-001 → UC-02 (Submit Bid on Load), UC-06 (Verify Carrier Credentials) • FR-002 → UC-01 (Post Freight Load) • FR-003 → UC-03 (Award Load to Carrier) • FR-004 → UC-05 (Submit Electronic Proof of Delivery) • FR-005 → UC-04 (Track Shipment Status) • NFR-001 → UC-07 (Release Escrow Milestone Payment) • NFR-002 → all use cases (platform-wide constraint)